

BHARATI PATIL

Pune, India | pbharati2002@gmail.com | +91 7020276545

GitHub: <https://github.com/bharati1802> LinkedIn: <https://linkedin.com/in/bharatipatil1802>

Objective

I am an AI fresher seeking an internship opportunity to apply my knowledge of Agentic AI, LLM-based applications, and IoT systems. I am interested in working on real-world AI projects and building skills in intelligent automation systems, multi-agent workflows, and AI-based applications while continuously learning and improving my industry-ready skills.

Technical Skills

- **Languages:** Python, Google Collab, Jupyter, Matplotlib
- **AI/ML:** Machine Learning, NLP, Deep Learning (Basics)
- **Generative AI :** Suno AI, Eleven Labs, DALL-E, Runway & Midjourney
- **Agentic AI:** Planner-Executor Architecture, AI Agents, Multi-Agent Systems
- **LLM Concepts:** Prompt Engineering, Tokenization, Embeddings, Transformers, Prediction Sampling
- **Advanced:** RAG, Fine-tuning (Basics), Hallucination Handling, Guardrails
- **Tools:** Streamlit, Ollama (Local LLM), GitHub, VS Code, Jupyter
- **Data & APIs:** JSON, API Integration, Vector DB (FAISS – basics)
- **IoT:** ESP32, Arduino, Sensors, Cloud Integration (ThingSpeak)

Projects

Agentic Healthcare AI Assistant (LLM + Ollama)

GitHub: <https://github.com/bharati1802/AGENTIC-AI-HEALTHCARE-ASSISTANT>

- Built a multi-agent AI system using Planner-Executor architecture for healthcare claim automation
- Implemented JSON-based decision-making for tool selection and execution
- Integrated local LLM (Ollama - Llama3) enabling offline inference
- Developed interactive UI using Streamlit for real-time query handling
- Designed a mock API system to fetch claim status dynamically
- Stored user interactions using JSON-based data storage
- **Tech:** Python, Ollama, Streamlit, JSON

Student Management System

GitHub: <https://github.com/bharati1802/StudentManagementSystem>

- **Full-Stack Architecture:** Developed a CRUD-based Student Management System using **FastAPI** (Backend) and **Streamlit** (Frontend).
- **Database Management:** Integrated **MySQL** with **SQLAlchemy ORM** for efficient data persistence and schema management.
- **RESTful APIs:** Designed and tested secure API endpoints for real-time data handling, validated via **Swagger UI**.
- **Data Connectivity:** Implemented seamless communication between the UI and DB using the **Python Requests** library and JSON.

Smart-Grid Monitoring System

GitHub: https://github.com/bharati1802/Python_IoT-project

- Built a real-time IoT-based Smart Grid monitoring system using ESP32 and Python
- Developed fault detection modules to identify overheating, power theft, pole tilt, and wire cuts
- Integrated ThingSpeak cloud platform for real-time data visualization and automated alerts
- Developed IoT applications using Flutter (mobile) and Streamlit (web).
- Gain exposure to robotics, simulation, and drone systems.

Internship Experience

IoT Intern – Incubators Systems Pvt. Ltd., Pune

Jan 2026 – Feb 2026

- Worked with sensors and actuators using ESP32, Arduino, and Raspberry Pi.
- Implemented real-time device communication and networking protocols.
- Integrated cloud services for data acquisition, storage, and remote monitoring.
- Proposed an AI-based roadmap for predictive fault detection and system automation
- Collaborated with cross-functional team members to design, test, and troubleshoot IoT solutions

Data Science, ML & GenAI Intern – Incubators Systems Pvt. Ltd., Pune

Sep 2025-Ongoing

- Working on Data Science and Machine Learning concepts using Python and SQL
- Applying Natural Language Processing (NLP) techniques for text analysis and data processing
- Implementing Generative AI tools for coding assistance, automation, and productivity tasks
- Using AI tools for image generation, video generation, research, and content creation
- Exploring AI-based workflows for automation, presentations, and technical documentation

Core AI Knowledge

- **How LLM Works:**
Transformer-based architecture using attention mechanisms to process tokens and generate context-aware responses
- **Prompt Engineering:**
Zero-shot prompting and Few-shot prompting
- **RAG:**
Combines retrieval systems with LLMs for accurate, context-based responses
- **Fine-tuning:**
Adapting pre-trained models for domain-specific tasks
- **Hallucination Handling:**
Reducing incorrect outputs using structured prompts and grounding techniques

Education

- SSC – Maharashtra Vidyalaya, Barshi, Solapur (2018)
- HSC Maharashtra Higher Secondary Vidyalaya Barshi, Solapur (2020)
- Master of Computer Applications (MCA)
Sharda University, Greater Noida (2024 – 2026)
- Bachelor of Computer Applications (BCA)
Bharati Vidyapeeth, Pune (2020 – 2023)

Achievements & Activities

- Participated in Mega Ideathon 2026
- SuperX IoT Exhibition (Core2Web)
- AI & Placement MasterClass Seminar
- Marketing Intern – Core2Web Ambassador Program

